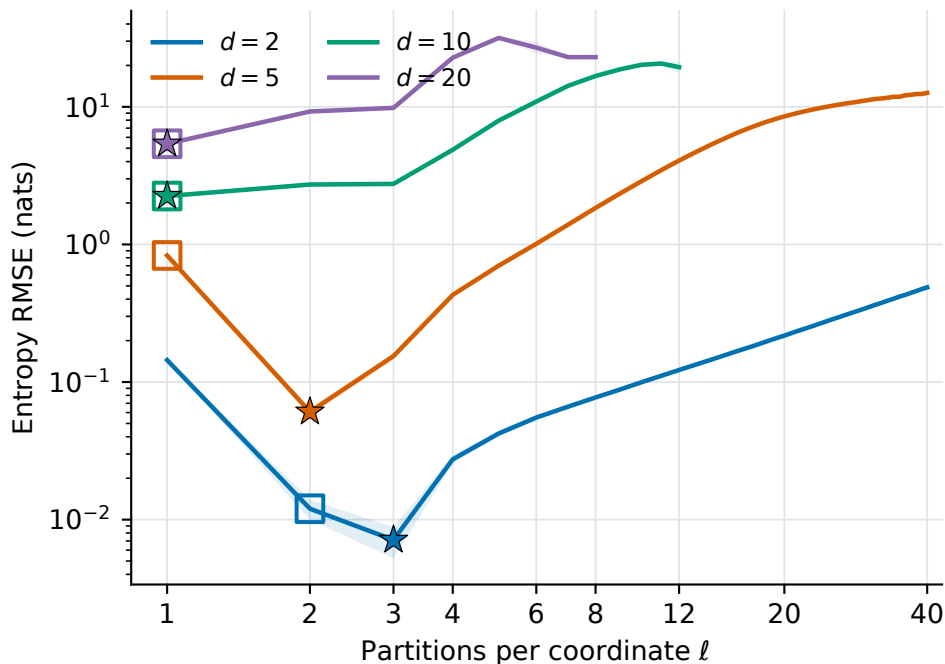


Gaussian data: empirical versus bound-based partition choice

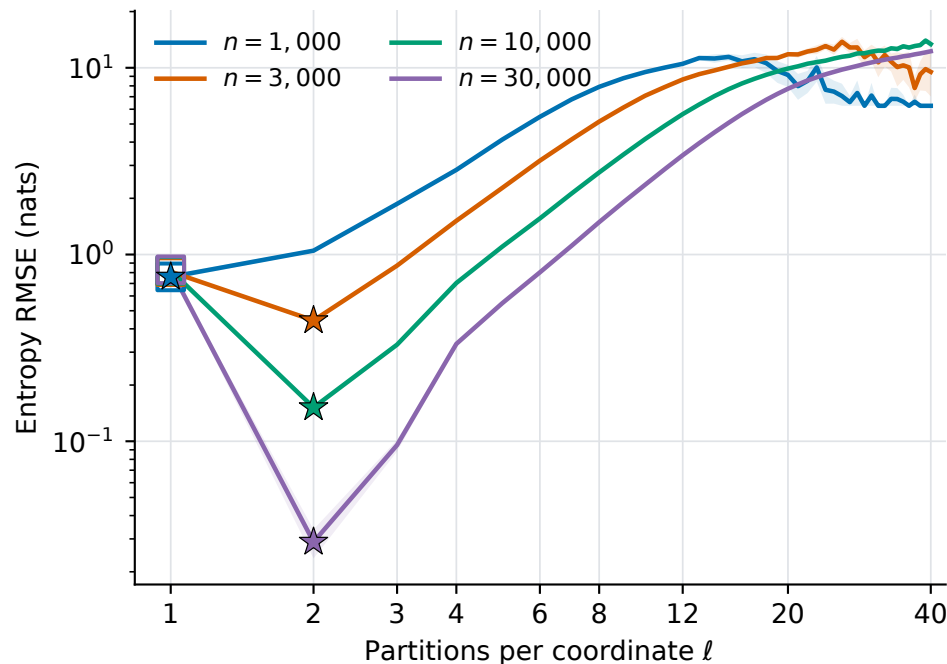
$X \sim N_d(0, \Sigma)$, $\Sigma_{jj} = 1$, $\Sigma_{ij} = 0.5$ ($i \neq j$); 30 repetitions per condition

★ Empirical RMSE minimum □ Bound minimum ($\delta = 0.05$)

(a) Fixed $n = 20,000$; varying dimension



(b) Fixed $d = 5$; varying sample size



Shading: pointwise 95% bootstrap intervals. All saved levels and repetitions are retained; no coverage filter.

The rounded dimension-aware rate with $C=1$ selects $\ell=1$ in all eight conditions. Stars are same-data grid minima.

Gaussian densities are outside the bounded-support theorem assumptions; minimizing its bound need not minimize RMSE.